

When two rooms share one AHU, local control is no longer enough.

EcoHVAC Guardian coordinates two simulated room twins, explains every constrained grant, and keeps predictive maintenance advisory until evidence and governance gates pass.

2

room twins

1

shared AHU

0

autonomous work orders

VALUE THESIS

Expose contention before it becomes hidden comfort debt.

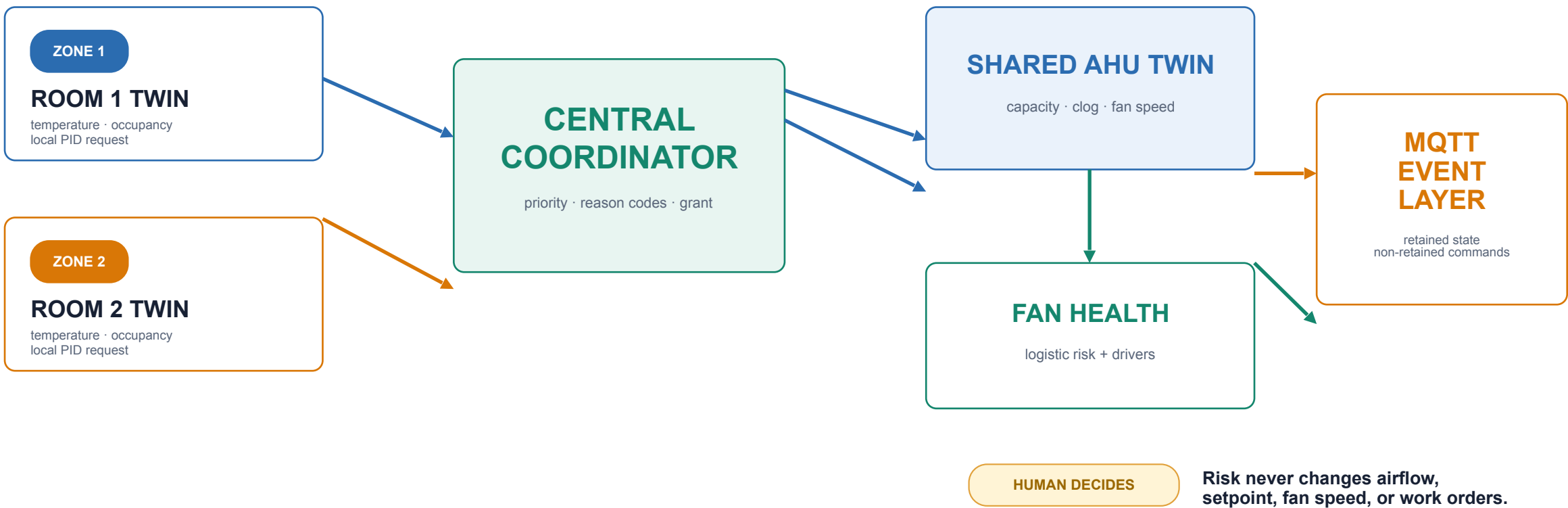
Prioritize transparently.

Measure energy and risk as connected consequences — not separate dashboards.

Scope boundary: simulated classroom pilot; no facility measurements, real ROI, production security, or physical autonomy.

Two local PID loops negotiate one finite shared-airflow budget.

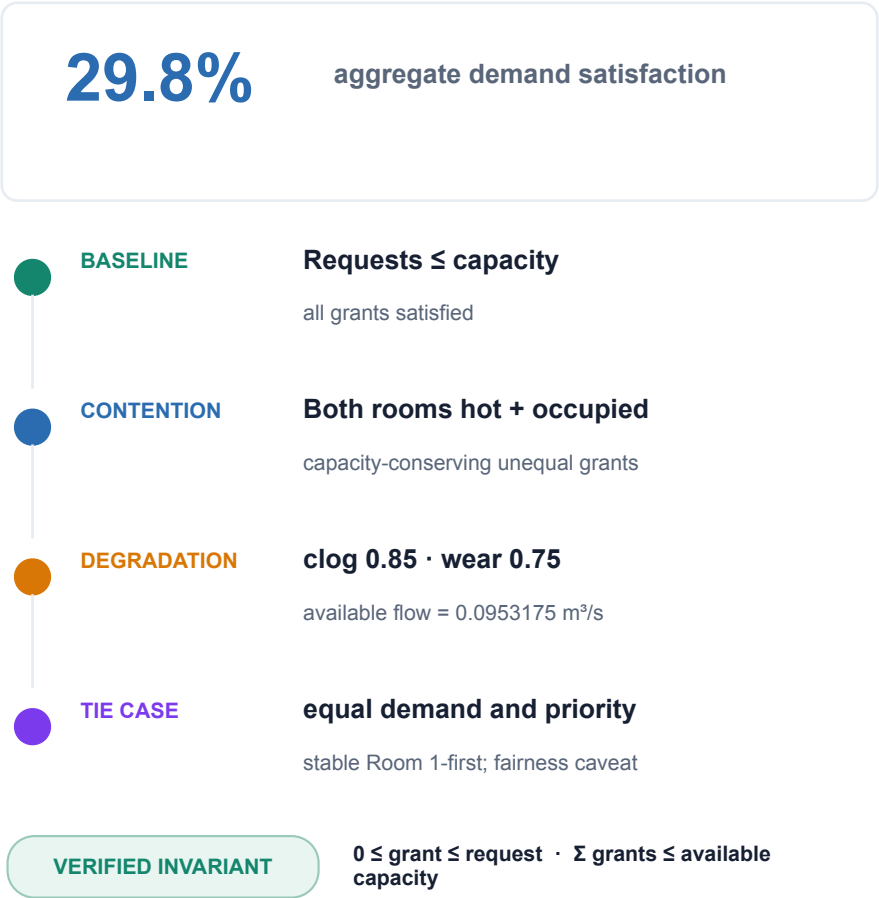
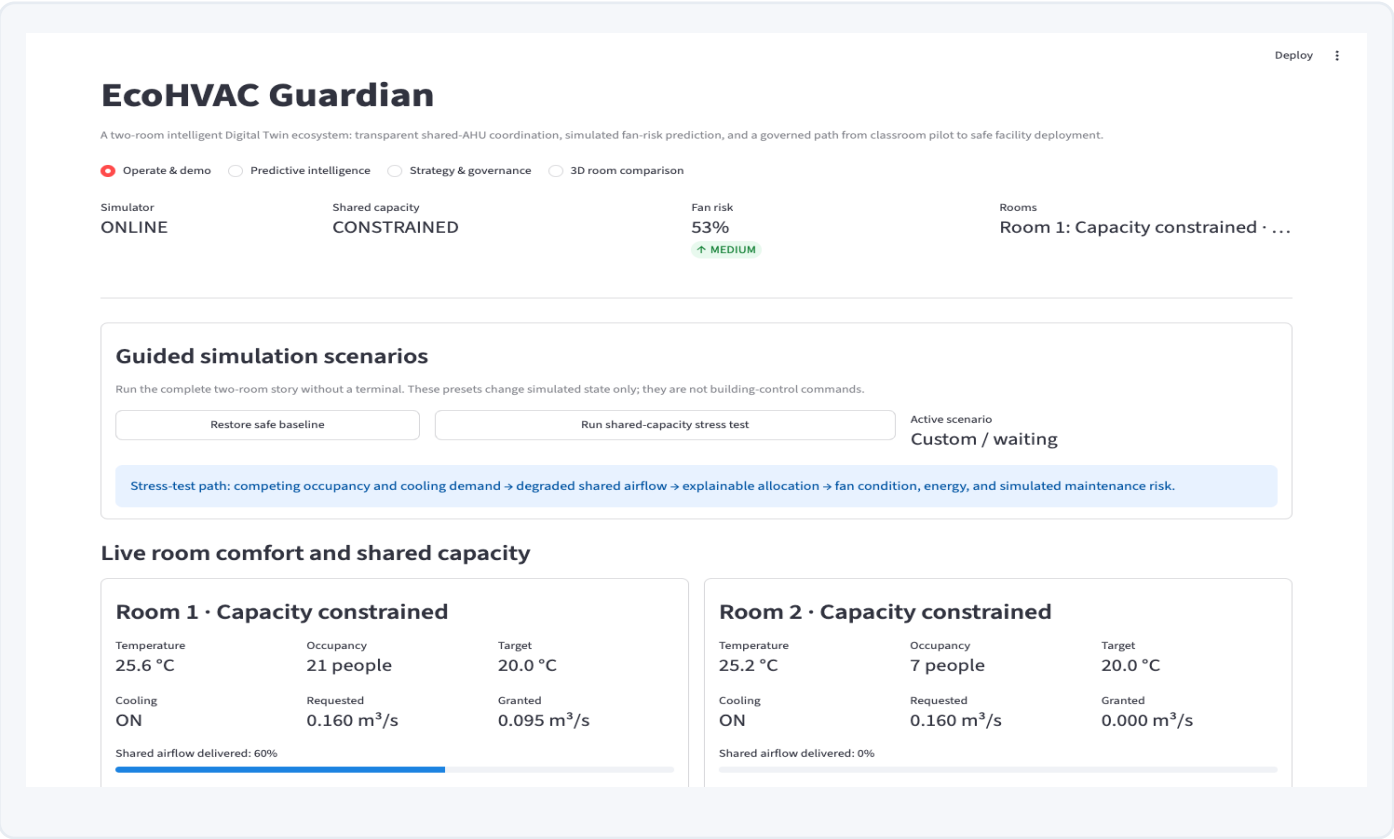
The central coordinator allocates grants; each room remains an independent thermal state, and fan risk remains advisory.



Source: architecture evidence figure; report Chapters 2, 4 and 5.

The stress test makes a hidden coupling visible: 0.320 m³/s requested, 0.095 m³/s available.

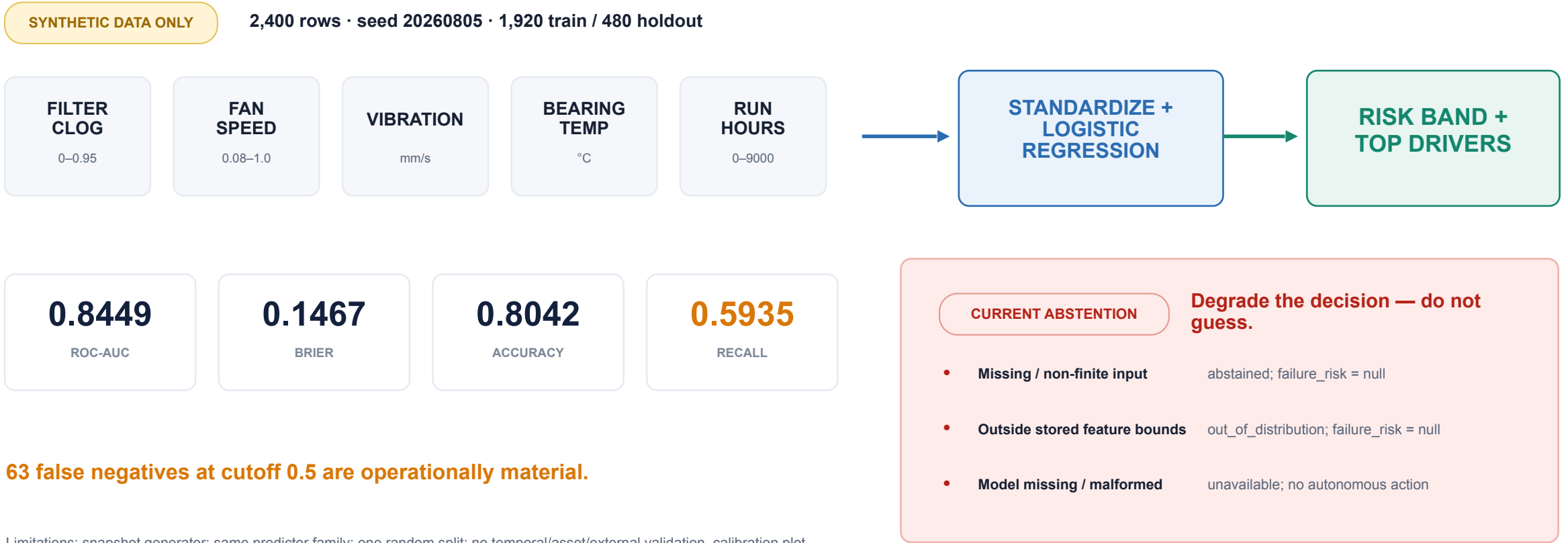
Current source implements atomic baseline and shared-capacity scenarios; the existing capture is retained as state evidence, not a repeatability certificate.



Source: current Project 2 stress capture; report §6.1–6.3.

The model recovers a seeded synthetic relationship — it does not estimate real-facility failure probability.

Five standardized inputs feed logistic regression; explanations are additive log-odds terms, not causal effects.

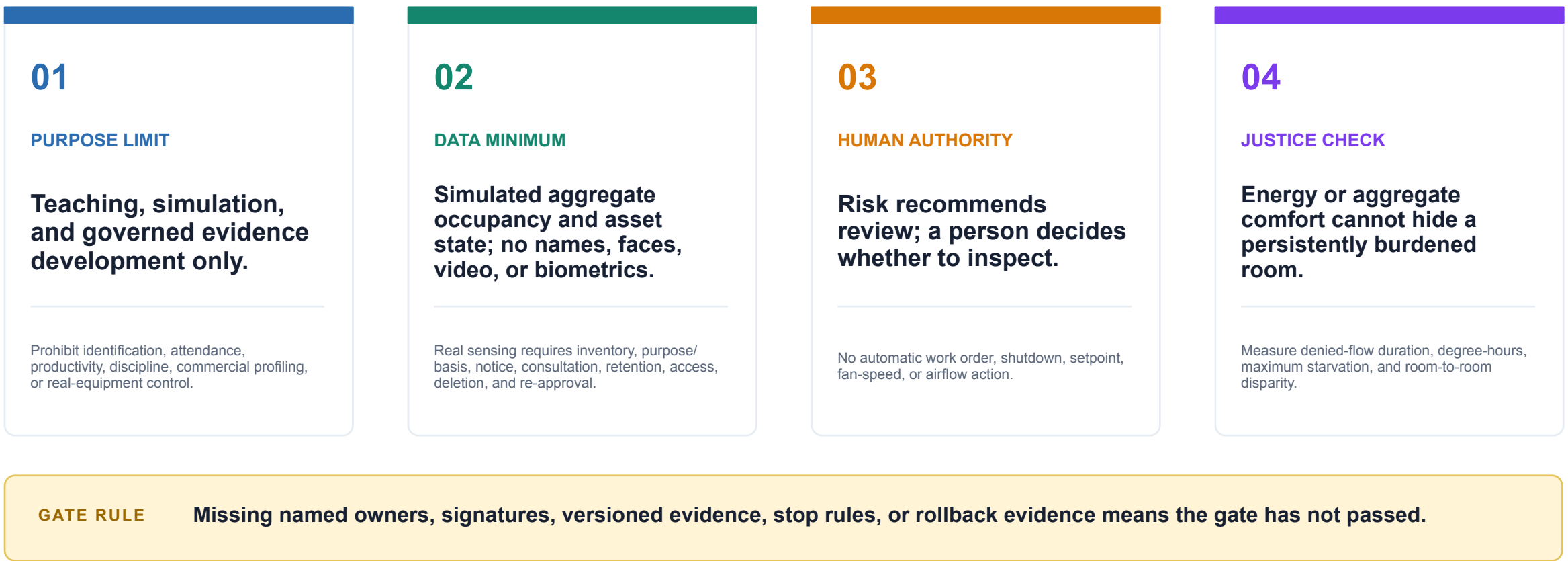


Limitations: snapshot generator; same predictor family; one random split; no temporal/asset/external validation, calibration plot, confidence intervals, multivariate uncertainty/drift monitoring, subgroup analysis, or cost-based threshold.

Source: stored metrics JSON; report Chapter 5 and model card.

The present ethical disposition is “pilot with restrictions,” not permission to automate a building.

Value must be reported with burden: denied airflow, comfort debt, false-alert workload, false negatives, privacy inference, and security exposure.



Source: report Chapter 7, ethics canvas and decision-rights matrix.

The classroom broker is anonymous and plaintext today; every production control shown here is a target.

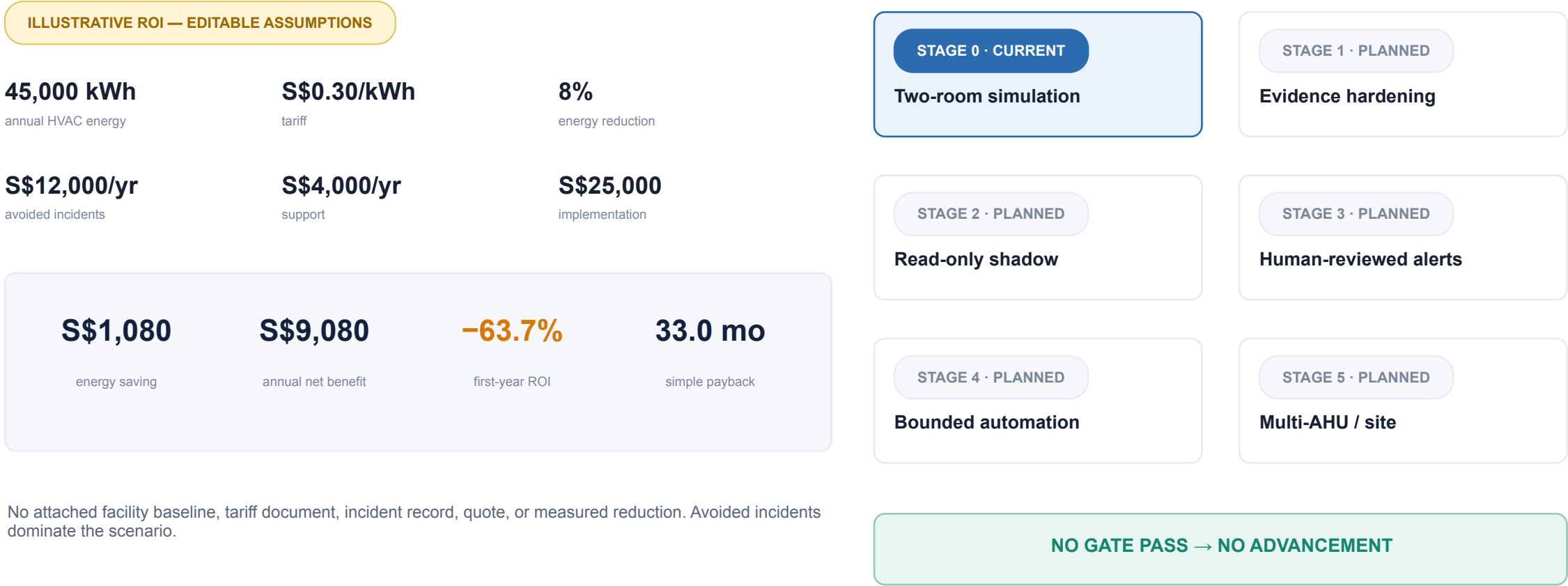
Current reliability checks are useful, but they do not authenticate actors or protect message confidentiality and integrity.



Source: mosquito.conf; report §7.7–7.9 and threat register.

Illustrative payback is 33.0 months — and it is not a savings claim.

Only Stage 0 is current. Every later stage starts only after evidence, security, safety, ethics, comfort, and value gates pass.



The current system demonstrates connected state and transparent decisions — not production readiness.

The next defensible move is evidence hardening, then a governed read-only shadow; do not jump to automation.

144

verified tests pass

JUnit: 144 tests · 0 failures · 0 errors · 0 skipped.

11/11

notebook code cells executed

Executed training notebook completed through nbconvert.

480

synthetic holdout rows

One seeded random split; no temporal, asset-group, external, or prospective validation.

KEY LIMITATIONS

- Archive a fresh current UI capture tied to the scenario transcript and evidence hashes.
- Single stress screenshot does not establish recovery or temporal fairness.
- 3D is read-only, local-only, and lacks a disconnected indicator.
- No production TLS/ACL, calibration/drift, field validation, or browser-level longitudinal evidence.
- No facility measurement, real ROI, production approval, or named gate authority.

PILOT ASK

Tie a fresh current UI capture to the archived scenario transcript, close remaining evidence gaps, and prepare the Stage 2 read-only security/data plan.

Safe, explainable, measurable coordination — proven one gate at a time.

Source: [release-evidence/project2](#) JUnit, notebook console, broker transcript; report validity/roadmap.